

[1] predicted student dropout using activity logs from the Moodle platform. They applied the CatBoost model and used a balancing method called ADASYN to fix data problems. Their final model performed very well, achieving an F1-score of 0.86. The study found that long periods of student inactivity on Moodle was the biggest warning sign of dropout risk.

[2] Studied the problem of student dropout in the higher education system of Finland. In this research, the authors used both traditional academic data and Moodle (LMS) activity records. They selected main features like total completed credits, changes in GPA, and course failure history. For classification, they applied machine learning models such as CatBoost, Random Forest, and **XGBoost**, which achieved high accuracy. However, the major limitation is that these advanced models work like a "**black-box**." They can predict who will drop out, but they do not explain the exact reasons for individual students. Additionally, since the study used data from a Western educational system, the findings may not directly apply to universities with different grading frameworks.

[3] predicted student dropout at a Hungarian university using interpretable machine learning on a dataset of 8,508 students. Using pre-enrollment data like high school grades, their CatBoost model achieved a 0.774 AUC score. They used SHAP and LIME to explain that high school GPA and math scores were the top predictors of student success.

[4] grouped at-risk students by analyzing Moodle activity logs. They used PCA and UMAP for data reduction, followed by clustering tools like BIRCH and DBSCAN to group students by engagement. Using Random Forest, they found that the BIRCH model worked best and showed that students with low online interaction had a dropout risk of over 61%

Research Gap and Our Contributions

To solve the limitations found in past studies, our research brings some new ideas. While older works like [1] and [2] rely strictly on "black-box" models, newer studies like [3] and [4] have started using Explainable AI (XAI) tools like **SHAP**. However, these newer studies still face major gaps: [3] strictly relies on data from *before* university enrollment, and [4] uses an extremely small dataset of only 152 students from a single course. Our study introduces major improvements to fix all these limitations together. First, we provide transparent, student-level risk factors using SHAP. Second, we combine both university institutional records (CGPA progression) and ongoing LMS logs on a much larger dataset, which past works missed. Third, instead of just looking at final

results, we use **temporal modeling** to find the earliest time to help a student. Finally, this project uses a real-world dataset from the **CSE department of BUBT**, making it perfectly tailored for the Bangladeshi private university context.

References

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